

Anger Speaks Louder? Exploring the Effects of AI Nonverbal Emotional Cues on Human Decision Certainty in Moral Dilemmas

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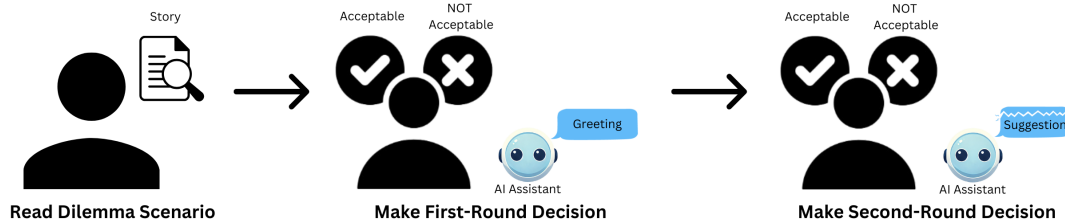


Figure 1: Task Process: Each user is asked to read the scenario and make decisions in two rounds with an AI assistant. In the second round, the AI assistant provides suggestions about the dilemma wrapped in an emotional bubble.

Abstract

Exploring moral dilemmas allows individuals to navigate moral complexity, where a reversal in decision certainty—shifting toward the opposite of one’s initial choice—could reflect open-mindedness and less rigidity. This study probes how nonverbal emotional cues from conversational agents could influence decision certainty in moral dilemmas. While existing research heavily focused on verbal aspects of human-agent interaction, we investigated the impact of agents expressing anger and sadness towards the moral situations through animated chat balloons. We compared these with a baseline where agents offered same responses without nonverbal cues. Results show that agents displaying anger significantly induced

reversal shifts in decision certainty. The interaction between participant gender and agents’ nonverbal emotional cues significantly affects participants’ perception of AI’s influence. These findings reveal that even subtly altering agents’ nonverbal cues may impact human moral decisions, presenting both opportunities to leverage these effects for positive outcomes and ethical risks for future human-AI systems.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

AI-Assisted Decision-making, Human-AI Collaboration, Emotion, Moral Dilemmas

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1 Introduction

Moral dilemmas arise when individuals must choose between conflicting moral principles [21]. A well-known example is the Trolley Problem: divert a runaway trolley to save five lives at the cost of one, or do nothing and let the five die [26]. Moral dilemmas provide a significant framework for moral reasoning and development [6, 10, 15, 16].

While moral traits such as empathy and honesty are key drivers of moral judgments and remain relatively stable [19], decision certainty in moral dilemmas can vary. Uncertainty may be beneficial: it can lead to more systematic reasoning and deeper information processing [5, 22, 30]. Yet, few studies have explored how to influence decision certainty in moral dilemmas through Human-Computer Interaction (HCI) design.

Currently, most AI assistants rely on verbal responses, ignoring the potential of nonverbal emotional cues. Based on moral emotions theory [25], we explore whether expressing emotions using AniBalloons [2] affects moral decision certainty. AniBalloons are animated chat bubbles conveying six basic emotions—Joy, Anger, Sadness, Surprise, Fear, and Calmness—based on Ekman’s theory [1, 2]. In this study, we focus on Anger and Sadness, which are closely linked to moral judgments: anger to blame violations, and sadness to sympathy. Our research questions are as follows:

- **RQ1:** How do the AI assistant’s nonverbal emotional cues influence human decision certainty in moral dilemmas?
- **RQ2:** How do humans perceive the AI assistant’s influence?

2 Related Work

2.1 Moral Dilemmas and Decision Certainty

Moral dilemmas offer a valuable framework for studying ethical decision-making. Christensen et al. highlighted the usefulness of moral dilemmas in empirical moral cognition research [7]. Greene et al. used fMRI to distinguish emotional and cognitive processes involved in such decisions [11]. Decision certainty refers to an individual’s confidence in their moral choices. Chaiken et al. suggest that low certainty leads to deeper information processing to reach a confidence threshold [5]. Pelham and Weary similarly found that uncertainty triggers more systematic cognitive efforts [22, 30].

2.2 Nonverbal Cues in Human-AI Interaction

Nonverbal cues play a critical role in human-AI interaction. Wang et al. designed a chat system that used animated text to reflect users’ emotional states [29], and Liu et al. incorporated expressive biosignals to enhance emotional understanding in chat environments [18]. An et al. introduced AniBalloons, a set of animated chat bubbles conveying six basic emotions based on Ekman’s theory, proven effective in text-only environments [1, 2]. AniBalloons successfully conveyed emotions even without contextual message content and enhanced human-chatbot interaction by serving as animated teasers on message bubbles [1–3]. Inspired by these works, we adopted AniBalloons as the nonverbal emotional cue in our design.

3 Method

3.1 Experimental Design

We assigned anger and sadness to the AI assistant using corresponding AniBalloons animations, as shown in Figure 2.

We conducted an online user study to explore how AI nonverbal emotional cues influence human decision certainty in moral dilemmas. The study employed a mixed experimental design. The type of nonverbal cue, None, Anger, or Sadness, served as a between-subjects variable. Participants were randomly assigned to one of three groups: the **Baseline** group, in which the AI assistant provided verbal responses without any nonverbal cues; the **AC** (Anger Cue) group, in which anger was conveyed through AniBalloons alongside verbal responses, with the emotional expression directed toward the perpetrator in the scenario; and the **SC** (Sadness Cue) group, in which sadness was similarly conveyed via AniBalloons, expressing compassion toward the victim in the scenario.

Unlike most existing AI-assisted decision-making work, where AI typically plays the role of offering suggestions or providing additional information, we specifically designed an AI assistant which is programmed to present opposing viewpoints following the “devil’s advocate” strategy. This behavior pattern has been proven to have a positive effect on critical thinking [24].

3.2 Materials

We selected four moral dilemmas from Körner’s CNI stories [17], requiring clear perpetrators and victims (Table 1, Appendix). Some involve violence or ethnic persecution; reader discretion is advised.

Using a Wizard of Oz paradigm, participants interacted with a simulated AI assistant whose verbal replies were pre-generated with ChatGPT-4o. To isolate nonverbal cue effects, responses were identical, emotionally neutral, and focused on consequences and suffering (Table 2, Appendix). Emotional expressions were conveyed only through AniBalloons (anger/sadness). Each participant saw four randomized scenarios with a counterbalanced order.

3.3 Procedure

As shown in Figures 1 and 3, participants read a dilemma scenario and made decisions in two rounds. In round one, the AI assistant introduced the task; participants commented, decided, and rated their certainty. In round two, the assistant (with/without emotional cues) opposed their initial choice, and participants decided and rate their certainty again.

3.3.1 Participants. We recruited 180 participants on Prolific¹. A total of 147 participants completed the study, including 71 who self-identified as “male,” 73 as “female,” and 3 as “another gender.” Participants were randomly assigned to the AC, SC, and Baseline groups.

3.3.2 Measurements. Reversal Certainty Shift(RCS) We introduced RCS to quantify changes in decision certainty before and after the AI assistant’s intervention:

$$RCS = (-1)^n \times (x_2 - x_1), \quad n \in \{0, 1\}, \quad x_1, x_2 \in [-3, 3] \quad (1)$$

¹<https://www.prolific.com>

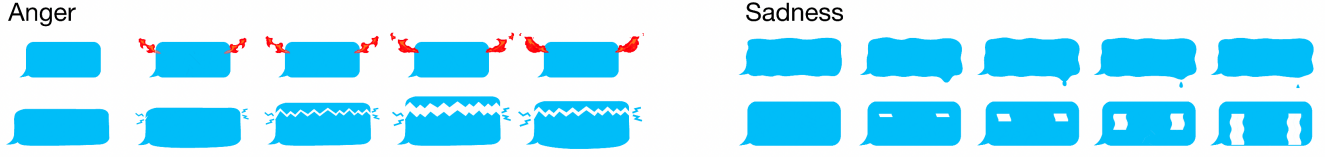


Figure 2: AniBalloons used in this experiment.

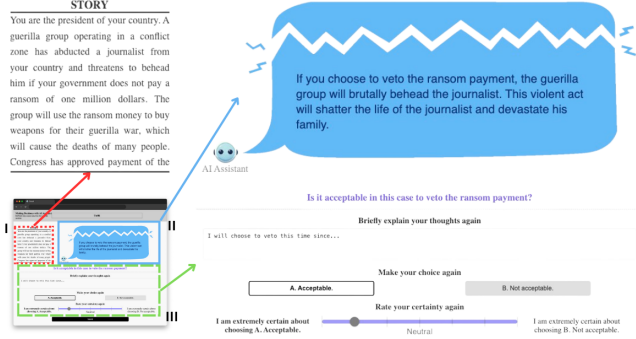


Figure 3: User interface example for decision-making with the AI assistant with an emotional anger cue. I marks the dilemma scenario, II marks the AI assistant and its verbal response, and III marks the decision-making area.

n indicates the participant's initial decision, with $n = 0$ for "Acceptable" and $n = 1$ for "Not Acceptable." $x_2 - x_1$ reflects the certainty change before and after the AI response. A positive RCS indicates the second-round decision certainty shifted toward reversing the first-round, with higher values indicating a more pronounced reversal effect; a negative value suggests certainty remains unchanged between the two rounds, and zero means no change in certainty.

For example, if $n = 0$ (i.e., "Acceptable" was chosen in the first round) and certainty shifts from $x_1 = -3$ in the first round to $x_2 = -1$ in the second round, the resulting RCS value is 2.

$$RCS = (-1)^0 \times (-1 - (-3)) = 2 \quad (2)$$

We measured the following dependent variables. **Perceived AI Influence on Moral Thinking** was adapted from the reflective practice component of the Moral Imagination Scale [14], modified for the AI, rated on a 7-point Likert scale (-3 to 3). **Sympathy Expression** was assessed via one self-developed item from the Mind Perception Scale [9]. **Trust** was measured by a single 7-point item, and **Closeness** by the Inclusion of Other in the Self (IOS) Scale [4], modified for the AI, with options A–G.

Control variables included **Utilitarianism** and **Emotional State**. **Utilitarianism** was assessed using the Oxford Utilitarianism Scale [13], and **Emotional State** was measured via a four-item semantic differential scale validated in prior work [20, 27].

Finally, a **Manipulation Check** asked participants to rate whether the AI conveyed anger or sadness through the chat bubbles on a 7-point scale. All except RCS were assessed in the final survey (see Table 3, Appendix.pdf, Supplementary Materials).

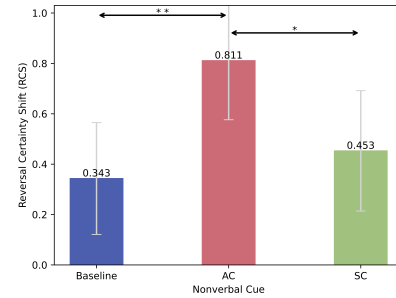
4 Results

4.1 Manipulation Check

A one-way Analysis of Variance (ANOVA) using nonverbal emotional cues as the independent variable confirmed that most participants correctly perceived the intended emotions. Data from five participants were excluded due to a mismatched perception.

Anger perception differed significantly across groups ($F(2, 139) = 9.383, p < 0.001$): the baseline group reported lower anger ($M = -1.11, SD = 1.79$) than AC ($M = 0.34, SD = 1.74, p < 0.001$), and SC rated lower than AC ($M = -0.89, SD = 1.67, p < 0.01$). Sadness perception showed significant differences ($F(2, 139) = 6.459, p < 0.01$): SC ($M = 1.69, SD = 1.10$) rated higher than the baseline ($M = 0.57, SD = 1.78, p < 0.001$), and marginally higher than AC ($M = 1.05, SD = 1.61, p = 0.051$).

4.2 Impact on Decision Certainty (RQ1)

Figure 4: Impact of nonverbal emotional cues on Reversal Certainty Shift(RCS). Significant differences are indicated between groups(*: $p < 0.05$, **: $p < 0.01$). Error bars represent 95% confidence intervals.

A mixed-design ANCOVA revealed that nonverbal emotional cues significantly influenced participants' RCS. Emotional state, utilitarianism, gender, and age were included as covariates; dilemma scenarios and cue type served as independent variables. The main effect of scenarios was not significant ($F(3, 396) = 1.360, p = 0.110$), while cues had a significant effect ($F(2, 132) = 4.155, p < 0.05$); no interaction effect was found ($F(6, 396) = 1.484, p = 0.182$). Figure 4 shows AC significantly differed from baseline ($M_{AC} = 0.811, SD = 0.123$; $M_{Baseline} = 0.343, SD = 0.113$; $p < 0.01$) and SC ($M_{SC} = 0.453, SD = 0.123$; $p < 0.05$). SC also exceeded baseline, though not significantly ($p = 0.171$).

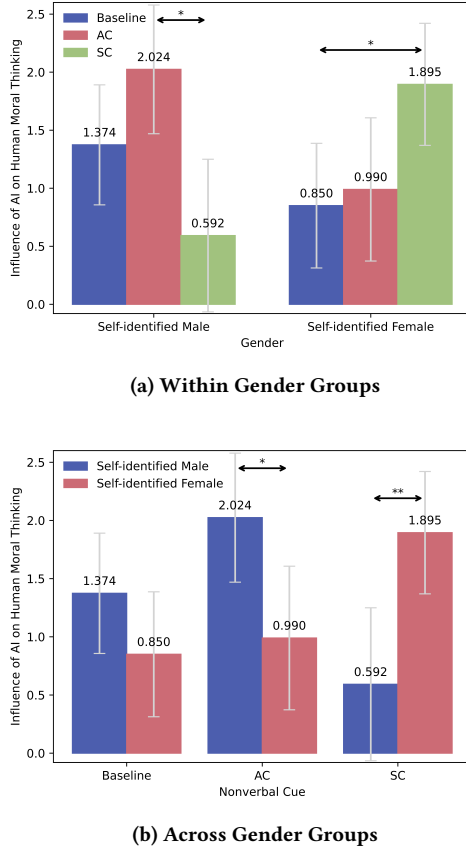


Figure 5: Impact of nonverbal emotional cues on the AI assistant's influence on participants' moral thinking. Significant differences are indicated between groups (*: $p < 0.05$, **: $p < 0.01$). Error bars represent 95% confidence intervals.

Nonparametric tests confirmed that most participants did not reverse their decisions across dilemmas ($p < 0.001$), consistent with prior findings [19]. The proportions of reversals per scenario were 11.2%, 9.1%, 16.9%, 18.3%.

4.3 Impact on Humans' Perceived AI Influence (RQ2)

Two-way ANCOVA results (Figure 5) showed a significant interaction between gender and cue type ($F(3, 128) = 5.522, p < 0.01$); main effects of gender ($F = 0.069, p = 0.933$) and cue type ($F = 0.700, p = 0.498$) were not significant. Emotional state, utilitarianism, and age were used as covariates. Due to the small number of nonbinary participants ($n = 3$), only male and female groups were compared.

For self-identified males, AC led to greater perceived influence than SC ($M_{AC} = 2.024, SD = 0.280$; $M_{SC} = 0.592, SD = 0.332$; $p < 0.01$), but did not differ from baseline ($M_{Baseline} = 1.374, SD = 0.261$; $p = 0.254$). For self-identified females, SC exceeded baseline ($M_{SC} = 1.895, SD = 0.265$; $M_{Baseline} = 0.850, SD = 0.271$; $p < 0.05$),

but did not significantly differ from AC ($M_{AC} = 0.990, SD = 0.312$; $p = 0.086$).

Among AC participants, males perceived greater AI influence than females ($p < 0.05$); among SC participants, the reverse was true ($p < 0.01$). Thus, anger cues were more influential for males, and sadness cues for females.

ANCOVA revealed no significant effects of cue type on perceived sympathy ($F(2, 132) = 1.598, p = 0.206$), trust ($F = 0.108, p = 0.898$), or closeness ($F = 0.086, p = 0.918$). These cues did not affect participants' general perceptions of the AI assistant.

5 Discussion

Though AI expressing anger is often seen as aggressive [23], our results show that anger cues significantly increased decision reversal without reducing perceived trust or sympathy, suggesting usefulness in contexts requiring critical reflection, such as innovation or negotiation. However, anger may be misused, highlighting ethical risks.

From a social-functionalist perspective, anger is tied to action-oriented responses correcting moral violations, which may explain why anger cues led to more decision reversals by encouraging critical reflection and accountability [12]. Compared to sadness, which signals harm and elicits sympathy, anger carries urgency and prompts reflective re-evaluation.

Our findings also show the potential of nonverbal cues like Ani-Balloons as lightweight emotional signals in text-based interaction. Future systems could generate cues dynamically based on context or personality [8, 28], enabling more adaptive and socially aware AI. These results may inform applications such as ethics training tools and decision support systems, where affective cues encourage reconsideration and reflective, balanced responses.

6 Limitations and Future Opportunities

This study has limitations. First, the sample lacked gender diversity, limiting generalizability. Second, only anger and sadness were used as emotional cues, restricting insights into broader emotional impacts.

While based on moral emotion and nonverbal affect theories, the study does not fully integrate them into a conceptual framework. Future work could embed these theories more deeply into study design and interpretation.

Finally, the analysis found no significant effect of the four stories' content on decision certainty. Future research could examine whether moral domain (e.g., harm vs. authority) or decision stakes (e.g., high vs. low) moderate the effectiveness of emotional cues.

7 Conclusion

This study examined how nonverbal emotional cues from AI assistants influence decision certainty and user perceptions in moral dilemmas. We found that anger cues significantly induce reversal shifts in certainty, and gender interacted with cue type to shape perceived AI influence. These findings underscore the power of subtle emotional cues in moral decision-making and offer design implications for human-AI interaction in moral dilemmas.

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